

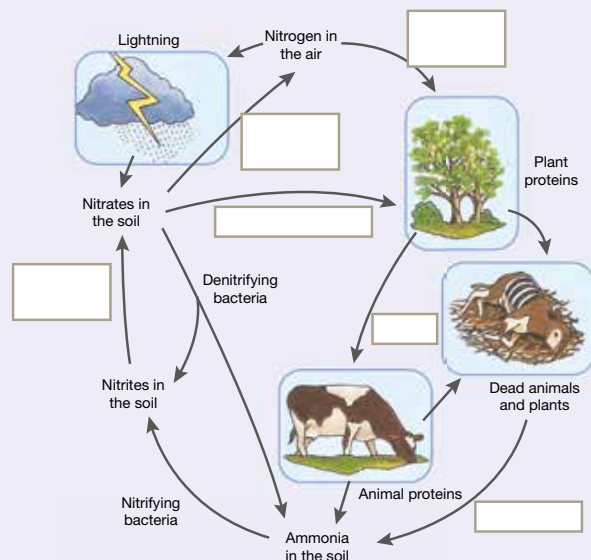
7.13 Review 1: Looking back

To answer questions online and to receive **immediate feedback** and **sample responses** for every question, go to your learnON title at www.jacplus.com.au. *Note:* Question numbers may vary slightly.

- Global warming is a current issue that is not going away.
 - Outline the most accepted view within the scientific community of the cause of global warming.
 - Describe examples of effects or consequences of global warming that have been suggested by scientists.
 - State your opinion about the possible (i) cause, (ii) effects and (iii) solutions for global warming.
 - View the top ten arguments about global warming that are used by sceptics. Rank these statements in order of most like your opinion to least like your opinion. Justify your ranking.
 - State the difference between an opinion, a theory and a fact.
 - Can scientists have opinions? If you agree, when, how and why should these be shared? If you do not agree, why not?
 - Should science play a part in the making of climate policy? Justify your response.
 - Suggest possible reasons for the climate debate.
- Demonstrate your understanding of the following groups of terms by using a visual Thinking tool to show the links between them.
 - Species, biodiversity, biodiversity loss, threatened, endangered, extinct, mass extinction
 - Biosphere, lithosphere, hydrosphere, biota, atmosphere, troposphere, stratosphere
 - Atoms, molecules, organelles, cells, multicellular organisms, species, population, ecosystem, biosphere
 - Stratosphere, climate change, greenhouse gas, fossil fuels, global warming, carbon dioxide, methane, nitrous oxide, biodiversity loss, enhanced greenhouse effect, cellular respiration, lithosphere
 - Carbon cycle, photosynthesis, cellular respiration, carbon dioxide
 - Water cycle, precipitation, transpiration, evaporation, hydrosphere
 - Ozone layer, ozone hole, CFCs, stratosphere
 - Abiotic factor, biotic factor, temperature, rainfall, climate, multicellular organism, ecosystem, biome
 - Greenhouse effect, enhanced greenhouse effect, global warming
- Constantly changing physical, chemical and biological cycles have contributed to the survival of various forms of life on Earth. Our life-support systems are not in good shape.

Using knowledge that you have gained from this chapter, comment on the statements above.

- What is meant by biodiversity and why is loss of biodiversity a concern?
- The mountain pygmy possum is restricted to an area of 6 km² in the Australian Alps. Suggest how such a restricted habitat may influence its chances of survival.
 - Suggest abiotic and biotic factors that may affect this possum.
 - Suggest how warmer temperatures and reduced snow may affect its lifestyle. Be specific in your response by including examples of different scenarios.
 - What is meant by the term *extinction*?
 - If this species was to become extinct, suggest implications for other organisms within its ecosystem.
- Copy the figure at right into your workbook and then use the following terms to complete the links: nitrifying bacteria, uptake by roots, denitrification, decomposition, feeding, nitrogen-fixing bacteria.

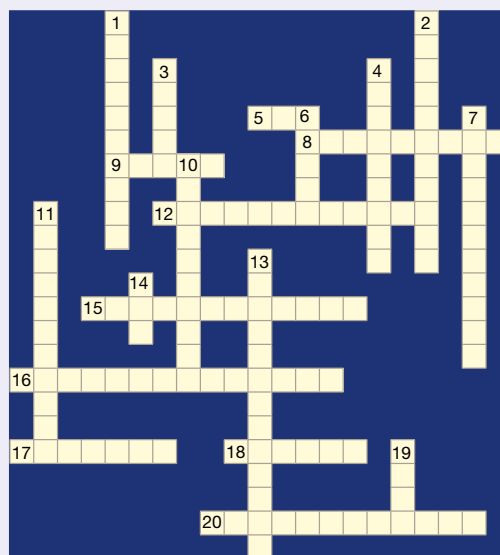


- Rising sea levels and saltwater intrusion associated with climate change are threats that Kakadu National Park is experiencing.
 - Suggest why these threats are associated with climate change.
 - Suggest effects that these new threats may have on the (i) biotic and (ii) abiotic parts of this ecosystem.
 - Suggest actions that could be taken to reduce the loss of biodiversity within Kakadu National Park.

8. Complete the crossword below.

ACROSS

5. Abbreviation of chlorofluorocarbon
8. Dynamic system of organisms interacting with each other and their environment
9. Planting these may help reduce the effect of global warming.
12. The ozone layer is located in this part of the Earth's atmosphere.
15. These bacteria convert nitrates in soil and water into nitrogen in the air.
16. Plants use this process to make glucose and oxygen.
17. An example of a greenhouse gas
18. Photosynthesis, respiration, death and decomposition are all processes within this cycle.
20. This term relates to the total variety of living things on Earth.



DOWN

1. A group of organisms of the same species in the same area
2. Includes water and dissolved carbon dioxide
3. A layer of this gas helps block out more than 95 per cent of ultraviolet rays entering the atmosphere.
4. The life-support system of our planet
6. Organisms are composed of these.
7. Global warming will lead to a rise in this factor.
10. The loss of a species from Earth
11. Includes rocks, coal and oil deposits, and humus in soil
13. This human activity can result in increased carbon dioxide levels in the atmosphere.
14. Abbreviation of deoxyribonucleic acid
19. Abbreviation of total ozone mapping spectrometer

9. Agriculture has had (and continues to have) a devastating effect on a number of marine ecosystems. Hypoxia in coastal zones from nitrogen and phosphorus outputs of agricultural and livestock industries is one such example.

- (a) Using your knowledge of the nitrogen and phosphorus cycles, explain how these outputs may damage marine ecosystems.
- (b) Suggest strategies that may reduce the negative impact that agriculture has on our ecosystems.

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